

October 5, 2026 04:01 AM GMT

Global Thematics & US Semiconductors | North America

What Does the Power Shortfall Mean for Semis?

We address key investor feedback from our power call: 1) impact to semis, 2) who we think can get power, and 3) potential upside drivers to our shortfall call. NVDA commentary confirms the power shortfall is real, but we remain convinced NVDA has line-of-sight into chip placement.

AlphaWise 

Key Takeaways

- Results of our recent deep dive into Powering AI: US data center developers face a >30% power shortfall through 2028.
- The power shortfall is real and affecting semis, but are convinced NVDA/AVGO can make 2027 numbers. Areas of concern: ASICs, memory, optics, other components.
- Capex is becoming more concentrated, with the hyperscaler/neocloud builds set to reach 60-95% of total global DC capacity additions from '26-28.
- We believe large, credit-worthy players will be able to energize (more bullish on hyperscalers), while less credit-worthy or int'l players may face issues.
- What could drive power upside? Faster BTM builds (turbines/fuel cells). Int'l expansion is happening but there's not enough capacity to absorb the US shortfall.


Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

We address the three most pressing investor questions on the power shortfall:

1) impact to semis, 2) who we think can get power, and 3) potential upside drivers that could close the shortfall. Last week, we published [Powering AI: Racking Up the Power Shortfall](#) where we sized the US data center power shortfall at 57 GW from 2026-28. We forecast this is 32 GW net of time to power solutions, which translates to a 34% net US power shortfall through 2028. Nvidia confirmed our thesis that there is not enough power to meet demand, and there are more semiconductors being built than there are places to put them. To be clear, in our view, this is not impacting Nvidia and Broadcom in 2027 (*more details below*), but we believe this could affect downstream components, like memory, optical, and other components like power management and analog parts in a bear case. We see a risk that ASICs will lose share in a power-constrained environment relative to GPUs,

MORGAN STANLEY & CO. LLC

Stephen C Byrd

Equity Analyst

Stephen.Byrd@morganstanley.com

+1 212 761-3865

Joseph Moore

Equity Analyst

Joseph.Moore@morganstanley.com

+1 212 761-7516

Brian Nowak, CFA

Equity Analyst

Brian.Nowak@morganstanley.com

+1 212 761-3365

Sabrina E Hao

Equity Analyst

Sabrina.Hao@morganstanley.com

+1 212 761-3315

Mason Wayne

Research Associate

Mason.Wayne@morganstanley.com

+1 212 761-6012

Gregory Gao

Research Associate

Greg.Gao@morganstanley.com

+1 212 296-3125

Nikhil Javeri

Research Associate

Nikhil.Javeri1@morganstanley.com

+1 212 761-3742

David Arcaro, CFA

Equity Analyst

David.Arcaro@morganstanley.com

+1 212 761-1817

Jeremy Herring

Research Associate

Jeremy.Herring@morganstanley.com

+1 212 761-4182

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

given the lower token production per watt for ASICs. Beyond 2027, as the US land, power, shell (LPS) shortfall worsens (driven by a combination of People, Power and Politics - see recent reports [here](#) and [here](#)), we see heightened risk across the entire chip ecosystem in terms of securing the necessary data center volumes to place all chip volumes. This shortfall will likely increase the urgency, and premia paid, to secure **(A)** time to power solutions in the US (such as Behind the Meter, or BTM, generation, as well as repurposing the grid access owned by former Bitcoin mining companies - see ways to play this dynamic [here](#)) and **(B)** data center growth outside of the US. Regarding this latter dynamic, we see pockets of excellent data center growth, in locations including [Korea](#), [Singapore](#), [India](#), [Australia](#), [China](#), and [Europe](#).

We have fielded a large volume of questions from investors on how our power shortfall thesis flows through to customers and the supply chain. Below, we address the top three debates right now:

1. What does the power shortfall mean for semis and the rest of the AI value chain?
2. Who is able to get access to power?
3. Is there a path for upside drivers, like international expansion and more BTM coming online, to help narrow the power shortfall more quickly?

1. What does this mean for semis and the rest of the AI value chain?

We remain convinced in Nvidia and Broadcom's ability to meet their 12 month forward outlooks, given 1) conservative management guidance takes into account possible Land-Powered Shell (LPS) shortage, 2) the visibility management has into chip placement, 3) geographic expansion outside the US, and 4) NVDA's higher tokens per gigawatt (GW) vs. peers creates a competitive advantage in a world where land, power, and shell is constrained. We are seeing this in real time, where the biggest users of ASICs, which have lower cost per GW, face heightened risk of inadequate Land-Powered Shell access. To some degree, we may have already seen this at Broadcom this year – after guiding for FY27 AI revenue well above \$100bn in March, the company's most recent guidance from September was for \$115bn, which is much less upside than investors initially anticipated. We think this puts some pressure on AMD as well, but the warrant dynamics with their largest customers do significantly de-risk their estimates in our view.

Nvidia and Broadcom have given 12 month outlooks for ~\$800bn of combined AI revenue. We don't think that's at risk due to power limitations today, but it's likely to make upside more difficult. Nvidia and Broadcom have a combined market share in XPU's of ~90%, and given their scale set the tone for much of the AI semis ecosystem. At this point, we do not see power bottlenecks putting 2027 forecasts at risk given the high degree of coordination and planning required to plan GW scale projects. A fully assembled GB200 NVL72 weighs over 3,000lbs and has 600,000 parts, so the level of coordination required between the data centers, semis suppliers, and power supply chain is quite a bit higher than traditional server racks. Data center customers are doing everything they can to avoid the opportunity cost of having assets idle when they could be revenue generating. We think the

management teams at Nvidia and Broadcom have a clear line of sight with respect to both the upstream and downstream inputs to their demand. Recent commentary from both CEOs around their respective earnings illustrates the point.

- *"we want to be sure... It's a judged number based on not just supply of memory wafers, but supply and availability and readiness of power sites" - Broadcom CEO Hock Tan: Sept 8, 2026*
- *"We work with power generator companies...We work with land, power and shell companies all around the world. And that helps prepare all of this computing that's going to be built.... And so, we just have a lot greater visibility now upstream and downstream" - Nvidia CEO Jensen Huang Aug 26, 2026*
- *"we're tracking every single gigawatt of land power shell around the world, literally everything on the planet." - Nvidia CEO Jensen Huang Sept 10, 2026*

We are also not just taking management's word at face value, and continue to hear from our industry contacts that securing LPS is the #1 priority for compute and hyperscaler management teams, and we do see some companies finding more success than others. Timing is also a factor - semiconductor companies recognize revenue when chips go to into the server assembly which can take several weeks to convert into complete systems, which can then take several weeks to be shipped, delivered, and installed.

Where would we be more concerned within semis if we do see a slowdown? If there's chip capacity expected to come online that cannot be deployed at data centers, we could see customers push out delivery or start canceling orders. There could be some bias to hold inventory of GPUs that can't be powered up to avoid losing positions in the queue, but given the annual cadence of new chip releases and limited power, the opportunity cost of using a less performant chip is high and companies should have the lead times to reroute orders to customers who are ready to take delivery. But in the event of a pause, we're likely to see the most disruption at the tail of the bullwhip, and where shortage driven pricing dynamics are supporting revenue and EPS. To us, that looks like memory, optics, and other small dollar content components that are most likely to be over-ordered such as power management and analog parts.

The power risk drives our preference for Nvidia, who has the broadest geographic and customer footprint and whose GPUs produce the greater tokens per watt of power access. Half of Nvidia's business is outside of the traditional hyperscalers, with companies like Neoclouds, sovereigns, and enterprises where their desire for compute fungibility is much higher than hyperscalers, and where they benefit the most from the scale of Nvidia's ecosystem support and full stack offering. Nvidia has 80 cloud partners listed on their website, of which 55 are outside the US (although they may have datacenters in multiple regions).

2. Who is able to get access to power, and who is at risk of being power-constrained?

We believe we're in an environment where data center energization is being concentrated across a handful of large, credit-worthy players. Feedback we have

received is that companies with credit support, revenue visibility and large volume needs – like the hyperscalers – are getting access to power more quickly, leaving smaller, less credit-worthy, or international players at risk. Categories of companies potentially at risk of inadequate US power access include smaller neoclouds lacking strong credit backing, non-US players, and chips that have a lower production of tokens per watt of power access (such as ASICs). Our Internet team led by Brian Nowak forecasts capacity additions from the hyperscalers and neoclouds (AMZN, GOOGL, MSFT, META, SPCX, CRWV, NBIS) at 60-95% of our forecasted global data center capacity coming online, after taking into account the US power shortfall ([Exhibit 1](#); [Exhibit 2](#)). We believe this could potentially "squeeze out" or put pressure on other players over time. We would also note that **1)** we're hearing Amazon is well-positioned to get access to power and **2)** GOOGL is adding the most incremental capacity in '27/'28 and thus could mathematically face more challenges in securing power for the compute they are buying.

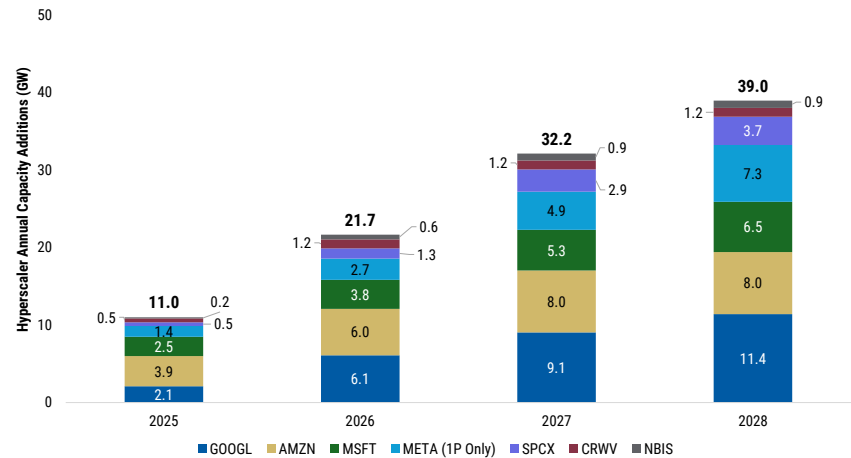
Exhibit 1: We forecast the hyperscalers and neoclouds representing ~60-95% of the global data center capacity we expect to come online each year.

Global Data Center Power Demand				
Year	2026	2027	2028	2029
Incremental Global DC Demand (GW)	32	56	86	124
(-) US Power Shortfall	(9)	(5)	(18)	(35)
Global DC Energized (GW)	23	51	68	89
Hyperscaler/Neocloud Capacity Additions (MSe)	22	32	39	
Hyperscaler/Neocloud % Global Capacity Additions	94%	64%	57%	

*Includes AMZN, GOOGL, MSFT, META, SPCX, CRWV, NBIS

Source: Morgan Stanley Research

Exhibit 2: Our Internet team forecasts the hyperscalers/neoclouds bringing on 32 GW in '27 and 39 GW in '28, with Google leading in terms of incremental capacity brought online.



Source: Morgan Stanley Research

3. Is there a path for upside drivers, like international expansion and more BTM generation, to help narrow the power shortfall more quickly?

While more data center capacity is shifting abroad, we don't believe there's enough capacity in ROW to absorb the US's net power shortfall (32 GW from '26-28). Companies are increasingly looking for capacity abroad, with Nvidia recently

noting that Australia is a region "with excess energy" where they stood up 2 GW for 2027. Given these latest data points, we move our US share of global compute to 55% from 60% previously (*more details below*). While we expect international expansion to continue, we don't see it as a solution to the US power problem. Conversations with our European data center analyst Emmet Kelly indicates limited room for additional spillover in Europe, given power constraints and the fact that planning permission is slow (see more [here](#)). While there could be potential for expansion into the Middle East, we believe this is difficult given geopolitical instability (see example of [AWS data center being hit by Iran strikes](#)). Our Asia data center team's forecast (led by Mayank Maheshwari) bakes in 14 GW of US hyperscaler demand spilling over into Asia through 2030 – and we still forecast a US power shortfall in spite of that – so any incremental spillover would require tapping into energy sources like coal. While there is a willingness to look abroad to build data centers, we don't believe there's enough capacity available to absorb the US power shortfall.

We do see potential upside to our forecast from behind the meter generation...

Net of our "time to power" solutions (BTM Turbines, Bloom Energy Fuel Cells, Nuclear capacity, and PSP Site Conversion - details [here](#)), we project that US data center developers will experience a 32 GW power shortfall through 2028, equal to 34% of power demanded. In these scenarios, we forecast 11-49 GW of BTM turbines/engines coming online, with 19 GW in our base case. *How do we get there?* 11 GW represents the firm, contracted BTM deals we see today, while 49 GW represents total possible production given factory capacity (keep in mind that labor availability, time required for complex engineering work, and permitting challenges limit this growth potential significantly). Our base case nets out announced projects; rather displaying *additional, behind the meter* deals signed through 2028. Our Utilities team, led by Dave Arcaro, can track 25-30 GW of manufacturer orders from 2026-28 and believes power developers can begin moving closer to adding 15 GW or more of BTM capacity per year beginning in 2028, narrowing the gap between OEM supply and annual operational capacity. Between our bull and base case, if 30 GW of BTM power successfully comes online from 2026-28, that implies a 21 GW shortfall.

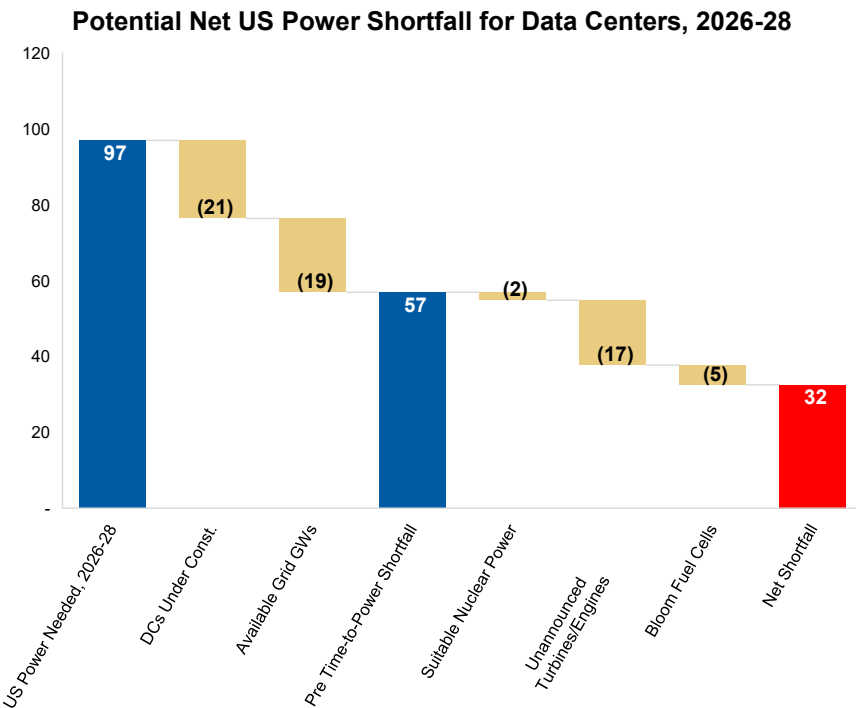
In addition, there could be potential upside to Bloom Energy fuel cells – we currently forecast 6GW of fuels cells from 2026-28, but there could be upside to this, given recent reports of site expansion (see team's thoughts [here](#)). Bloom Energy is a key beneficiary of this "time to power" shortfall, in our view. In our view, the urgency to secure power being expressed by AI players is much greater than appreciated. For any "time to power" provider that has a viable path to providing data center developers with power in 2027 and 2028, we believe such providers will be able to enter into lucrative agreements.

...however, People (labor availability) and Politics (growing pushback against data centers, and AI more broadly) are key driving forces in the opposite direction. Even if we can increase capacity of behind the meter generation coming online, people are still a bottleneck. [CSIS estimates](#) that by 2030 the US requires 140K+ incremental skilled trades people beyond current capacity in a scenario like the Second Industrial Revolution. The existing labor base would likely support 10-20 GW of annual gas additions, considering both large power plants and onsite

projects. However, companies are heavily investing in their labor forces to grow these numbers, such that it could be possible to reach 50+ GW of annual gas additions by the end of the decade. We've also written extensively about the impact politics, moratoriums, and public perception are having on the build out (see more [here](#), [here](#)).

Our US power shortfall remains at 57GW, and moves to 32 GW net of time to power solutions. With this recent update, we increase both our historical and projected non-AI Data Center demand, aligning with our Data Center team. We simultaneously revise our expectation of US Compute share down to 55% from 60%; these changes effectively balance out, leaving our total 2026-2028 power shortfall unchanged. As a result of increased global DC power demand projections, we find a 57GW power shortfall before time to power solutions and a net shortfall of 32GW through 2028, 34% of total demanded power. This shortfall is driven by the increasingly power-hungry rack-level GPU architectures (NVL72 or 144 versus extant 8U systems).

Exhibit 3: Net of "time to power" solutions, we project that US data center developers will experience a 32 gigawatt shortfall through 2028—this is a result of intensifying challenges with respect to People, Power and Politics



Source: Morgan Stanley Research.

Exhibit 4: Net of innovative "time to power" solutions, we project that US data center developers will face a 32-42GW Shortfall through 2028, 34-43% of demanded power. Note that the PSP sites line is not included in reducing the power shortfall but are included in either "Data Centers Under Construction", "Utility Grid Access", or "Addtl BTM Turbines/Engines".

Total US Data Center Power Demand (GW), 2026-28	97
Less Data Centers Under Construction	(21)
Less Utility Grid Access	(19)
Power Shortfall Before New "Time to Power" Solutions	57

	Low End	Midpoint	High End	Probability of Success
Solution #1: Addtl BTM Turbines/Engines	11	19	49	90%
Solution #2: Bloom Energy Fuel Cells	5	6	8	90%
Solution #3: Site DC at Op't Nuclear Plant	1	3	8	75%
Solution #4: Convert PSP Sites (included in turbines or DCs under construction)	8	13	19	90%
Probability-Weighted "Time to Power" Solutions through 2028	15	24	57	
Net Shortfall Through 2028	(42)	(32)	None	
Shortfall as a % of 2026-28 US DC Deployments	43%	34%	0%	

We believe this outcome is most likely

Source: Morgan Stanley Research Estimates

Exhibit 5: We project a 9GW Net Shortfall in 2026, scaling to a cumulative 68GWs in 2029.

Part IID.1. Annualized Cumulative Shortfall				
Year	2026	2027	2028	2029
Power Needed	22	51	97	160
US DCs Under Construction	(7)	(17)	(21)	(22)
Available/Contractable US Grid Capacity	(3)	(7)	(19)	(36)
Shortfall Before Time to Power Solutions	12	28	57	102
Suitable Nuclear Power	(0)	(1)	(2)	(3)
Bloom Fuel Cell Potential	(1)	(2)	(5)	(9)
Unannounced BTM Turbines/Engines	(2)	(10)	(17)	(23)
Crypto Sites Available (counted in DCs Under Construction or Turbines)	(2)	(6)	(12)	(18)
Est. Midpoint Net Shortfall (Midpoint)	9	15	32	68

Part IID.2. Annualized Year-By-Year Shortfall				
Year	2026	2027	2028	2029
Power Needed	22	30	45	63
US DCs Under Construction	(7)	(10)	(4)	(1)
Available US Grid Capacity	(3)	(4)	(13)	(17)
Shortfall Before Time to Power Solutions	12	16	29	45
Suitable Nuclear Power	(0)	(1)	(1)	(1)
Bloom Fuel Cell Potential	(1)	(2)	(3)	(4)
Unannounced BTM Turbines/Engines	(2)	(8)	(7)	(6)
Crypto Sites Available (counted in DCs Under Construction or Turbines)	(2)	(4)	(6)	(6)
Est. Midpoint Net Shortfall (Midpoint)	9	5	18	35

Source: Morgan Stanley Research Estimates

Exhibits

Exhibit 6: Many CEOs – particularly across the tech supply chain – have stressed the importance of the power shortfall and LPS availability.

Key Recent Quotes on Power Shortfall
"But as we look forward the bottleneck is no longer just in a single chip or a single technology or a single capability. The bottleneck moves to power. And more specifically the entire power domain from grid to site to core." - On Semiconductor President of Power Solutions Group, Achyut Shah (Sept 16)
"There is no question land, power and shell is a problem ." - NVIDIA CEO, Jensen Huang (Sept 10)
"I'm not saying supply chips and memory are not constraints. They are constraints. But at least they are constraints we can work with and are more deterministic. What is less deterministic is a power site ...It's also construction. Construction includes local permitting." - Broadcom CEO, Hock Tan (Sept 10)
"There will be a significant electricity shortfall next year ." - Elon Musk stressing the importance of power at the G20 tech ministers meeting (Sept 1)
"The fact that data center capacity is now measured in gigawatts rather than flops and tops, underscores one of the most defining challenges of the AI era. Power availability has become the primary constraint to further AI progress ." - CEO Analog Devices, Vincent Roche (Aug 19)
"But the biggest limitation today is power and cooling ...There is one challenge we all need to overcome, not just for our industry, but for our society and our planet, and that is power. Every model, every workload, every agent depends on power because at its core" - CEO HPE, Antonio Neri (June 16)

Source: Samaya, Websites, Morgan Stanley Research

Important note regarding economic sanctions. This report references jurisdictions which may be the subject of economic sanctions. Readers are solely responsible for ensuring that their investment activities are carried out in compliance with applicable laws.

Valuation Methodology and Risks

Broadcom Inc. (AVGO.O)

We value AVGO at 28X CY2027e ModelWare EPS of \$18.04. This is roughly 26x non-GAAP EPS of \$19.63, which is broadly in-line or below their AI peer group.

Risks to Upside

- Stronger AI revenues
- Faster recovery in core semis businesses
- Realized VMware synergies

Risks to Downside

- Lose networking share to Nvidia (Mellanox)
- ASIC chips are uncompetitive; lose customers to competitors
- Execution on VMware acquisition

NVIDIA Corp. (NVDA.O)

~20 our MW CY27 EPS estimate of \$15.01, in-line with the broader market and a discount to compute semis peers (AMD/AVGO/INTC) as high marketshare and gross margins leave limited levers for multiple expansion in the near term.

Risks to Upside

- Growth in inference and training propel data center revenue
- Gaming sales accelerate as GPU based AI PCs gain traction
- Nvidia can recapture lost revenue in China

Risks to Downside

- AI end markets don't materialize as expected, customers sharply reduce GPU purchases
- AMD reemerges as a viable GPU competitor
- Cloud customers outside of Google are able to develop competitive custom hardware

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V.. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V. and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David Arcaro, CFA; Stephen C Byrd; Sabrina E Hao; Joseph Moore; Brian Nowak, CFA.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The analyst or strategist (or a household member) identified below owns the following securities (or related derivatives): Gregory Gao - Astera Labs Inc(common or preferred stock), NVIDIA Corp.(common or preferred stock); Sabrina E Hao - NVIDIA Corp.(common or preferred stock); Nikhil Javeri - SanDisk Corporation.(common or preferred stock), Wolfspeed, INC(common or preferred stock); Brian Nowak, CFA - Advanced Micro Devices(common or preferred stock), Micron Technology Inc.(common or preferred stock), NVIDIA Corp.(common or preferred stock), ON Semiconductor Corp.(common or preferred stock).

As of August 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Advanced Micro Devices, Aeva Technologies Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., Arm Holdings plc, Astera Labs Inc, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, Intel Corporation, IonQ Inc, Marvell Technology Group Ltd, Microchip Technology Inc., Micron Technology Inc., NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., PDF Solutions Inc, Qorvo Inc, Qualcomm Inc., SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., Skyworks Solutions Inc, Synopsys Inc., Texas Instruments, Wolfspeed, INC.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Advanced Micro Devices, Aeva Technologies Inc, Analog Devices Inc., Broadcom Inc., Cerebras Systems, GlobalFoundries Inc, Intel Corporation, NVIDIA Corp., ON Semiconductor Corp., PDF Solutions Inc, Quantinuum, Semtech Corp., SiTime Corporation.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Micro Devices, Aeva Technologies Inc, Allegro Microsystems Inc, Amkor Technology Inc, Analog Devices Inc., Broadcom Inc., Cerebras Systems, GlobalFoundries Inc, Intel Corporation, IonQ Inc, Micron Technology Inc., Navitas Semiconductor Corp, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., PDF Solutions Inc, Quantinuum, SanDisk Corporation., Semtech Corp., SiTime Corporation, Texas Instruments.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro Devices, Aeva Technologies Inc, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., Arm Holdings plc, Astera Labs Inc, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Intel Corporation, IonQ Inc, Marvell Technology Group Ltd, Microchip Technology Inc., Micron Technology Inc., Navitas Semiconductor Corp, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., PDF Solutions Inc, Qualcomm Inc., Quantinuum, SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., SiTime Corporation, Skyworks Solutions Inc, Synopsys Inc., Texas Instruments, Wolfspeed, INC.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Advanced Micro Devices, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Intel Corporation, Marvell Technology Group Ltd, Microchip Technology Inc., Micron Technology Inc., NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Qorvo Inc, Qualcomm Inc., SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., SiTime Corporation, Synopsys Inc., Texas Instruments.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro Devices, Aeva Technologies Inc, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., Arm Holdings plc, Astera Labs Inc, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Intel Corporation, IonQ Inc, Marvell Technology Group Ltd, Microchip Technology Inc., Micron Technology Inc., Navitas Semiconductor Corp, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., PDF Solutions Inc, Qualcomm Inc., Quantinuum, SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., SiTime Corporation, Skyworks Solutions Inc, Synopsys Inc., Texas Instruments, Wolfspeed, INC.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Advanced Micro Devices, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Intel Corporation, IonQ Inc, Marvell Technology Group Ltd, Microchip Technology Inc., Micron Technology Inc., NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Qorvo Inc, Qualcomm Inc., SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., SiTime Corporation, Synopsys Inc., Texas Instruments, Wolfspeed, INC.

Morgan Stanley & Co. LLC makes a market in the securities of Aeva Technologies Inc, Ambarella Inc.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

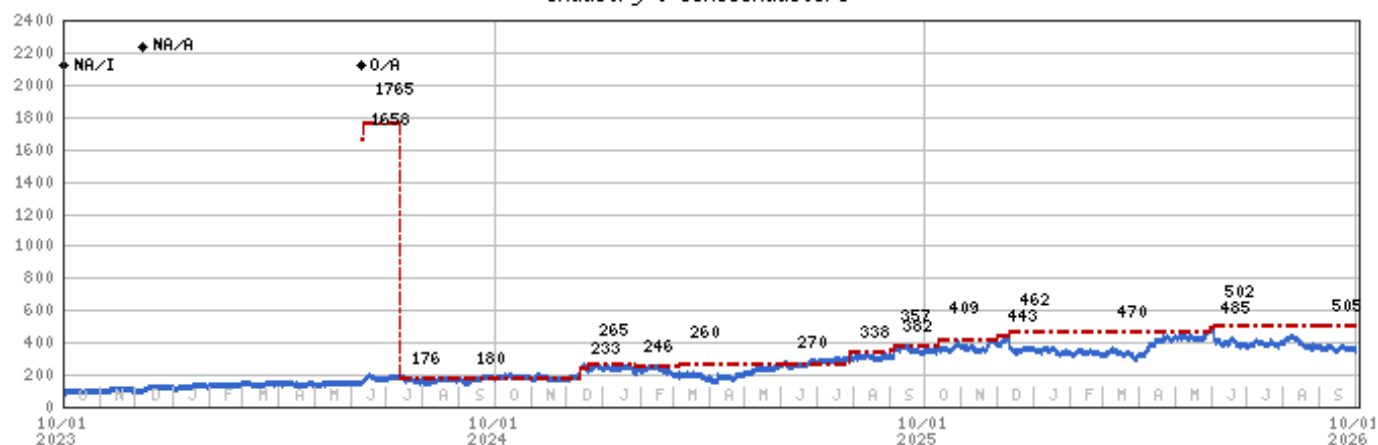
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Broadcom Inc. (AVGO.O) - As of 10/04/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 10/1/21 : O/I; 5/26/22 : NA/I; 10/10/22 : NA/I; 11/7/22 : NA/I; 12/7/23 : NA/A; 6/9/24 : O/A

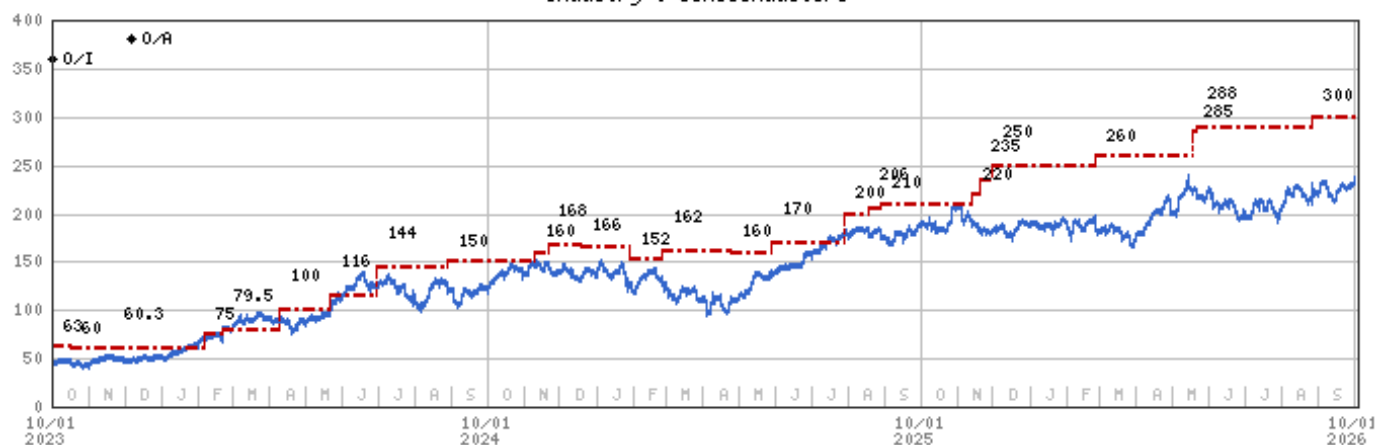
Price Target History: 9/2/21 : 572; 12/10/21 : 723; 3/3/22 : 703; 5/26/22 : NA; 10/10/22 : NA; 11/7/22 : NA; 6/9/24 : 1658;
6/12/24 : 1765; 7/12/24 : 176; 9/5/24 : 180; 12/13/24 : 233; 12/19/24 : 265; 1/27/25 : 246; 3/6/25 : 260; 6/6/25 : 270;
7/29/25 : 338; 9/1/25 : 357; 9/4/25 : 382; 10/13/25 : 409; 12/1/25 : 443; 12/12/25 : 462; 3/5/26 : 470; 5/31/26 : 485; 6/3/26 : 502;
9/2/26 : 505

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

NVIDIA Corp. (NVDA.O) - As of 10/04/26 GMT in USD Industry : Semiconductors



Stock Rating History: 10/1/21 : NA/I; 5/3/22 : E/I; 3/16/23 : 0/I; 12/7/23 : 0/A

Price Target History: 9/13/20 : NA; 5/3/22 : 21.7; 5/25/22 : 18.2; 11/17/22 : 17.5; 2/21/23 : 24.6; 2/22/23 : 25.5; 3/16/23 : 30.4; 5/24/23 : 45; 6/15/23 : 50; 8/24/23 : 63; 10/17/23 : 60; 11/21/23 : 60.3; 2/6/24 : 75; 2/21/24 : 79.5; 4/9/24 : 100; 5/22/24 : 116; 6/30/24 : 144; 8/29/24 : 150; 11/10/24 : 160; 11/21/24 : 168; 12/19/24 : 166; 1/29/25 : 152; 2/26/25 : 162; 4/25/25 : 160; 5/29/25 : 170; 7/29/25 : 200; 8/18/25 : 206; 8/28/25 : 210; 11/14/25 : 220; 11/20/25 : 235; 12/1/25 : 250; 2/26/26 : 260; 5/18/26 : 285; 5/21/26 : 288; 8/27/26 : 300

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

A member of Research who had or could have had access to the research prior to completion owns securities (or related derivatives) in the Analog Devices Inc., Broadcom Inc., Microchip Technology Inc., NVIDIA Corp., Qualcomm Inc.. This person is not a research analyst or a member of research analyst's household.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (<https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/>)

[MSICPL_Morgan_Stanley_Research_Audit_Report.pdf](#).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 1992062982) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by

FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/02/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	\$633.91
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	\$14.76
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	\$39.53
Ambarella Inc (AMBA.O)	O (03/29/2016)	\$68.65
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	\$56.02
Analog Devices Inc. (ADI.O)	O (11/16/2023)	\$417.15
Astera Labs Inc (ALAB.O)	O (05/11/2025)	\$350.33
Broadcom Inc. (AVGO.O)	O (06/09/2024)	\$355.14
Cerebras Systems (CBRS.O)	O (06/08/2026)	\$166.43
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	\$50.17
Intel Corporation (INTC.O)	E (02/22/2023)	\$119.33
IonQ Inc (IONQ.N)	E (04/25/2023)	\$43.77
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	\$272.29
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	\$81.32
Micron Technology Inc. (MU.O)	O (10/06/2025)	\$1,074.89
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	\$12.40
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	\$233.95
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	\$243.66
ON Semiconductor Corp. (ON.O)	++	\$84.89
Qorvo Inc (QRVO.O)	E (10/28/2025)	\$114.17
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	\$184.87
Quantinuum (QNT.O)	E (06/29/2026)	\$46.08
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	\$1,719.99
Semtech Corp. (SMTC.O)	E (04/06/2025)	\$194.88
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	\$222.76

SiTime Corporation (SITM.O)	O (09/16/2026)	\$709.44
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	\$85.03
Texas Instruments (TXN.O)	U (04/13/2020)	\$293.80
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	\$35.28
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	\$307.49
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	\$351.35
PDF Solutions Inc (PDFS.O)	O (09/08/2026)	\$55.50
Synopsys Inc. (SNPS.O)	O (09/08/2026)	\$489.90

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.